Hypothesis Stress-Test & Investment Decision Report v3.0.6.8

Powered by Vraimony
Protocol: ImmuneErrorRadar falsification protocol v3.0.6.9.7 | method=Vraimony methodology | posture=standard_followed_not_copied
Falsification receipt: id=ierfp-dd3472ce5e007a8c2a1c9cc4 | contentSha256=dd3472ce5e007a8c2a1c9cc4 | signedByVraimony=false

Documentation standard: ERF-inspired evidence receipt discipline: hash-first, portable, replayable, claim-bounded.
Citation line: This report follows the ImmuneErrorRadar falsification protocol, powered by Vraimony, based on the Vraimony evidence-receipt methodology.
Vraimony boundary: This receipt documents a falsification workflow and report payload hash; it is not medical advice, not clinical validation, not wet-lab evidence, and not proof of biology.
Report mode: verification-only | evidence-readiness receipt | investment-triage verdict | no discovery claim | no clinical advice

FINAL INVESTMENT DECISION v3.0.6.7: INVEST_IN_DATA_ONLY | claim=current independent/proof claim killed; downstream APM data question remains investable | killedClaim=independent_MYCN_adjusted_APM_driver_claim | criteria=kill:3/invest:2 | next=Invest only in data: full GPL5175 APM mapping, vector binding, GSE49710/TARGET-NBL direction replication, MYCN interaction/stability checks. No wet-lab yet. | probabilityReported=false
PREPRINT READINESS v3.0.6.8: PREPRINT_READY_WITH_DATA_ONLY_FOLLOW_UP | title=Hypothesis Stress-Test & Investment Decision Report v3.0.6.8 | killedAsWritten=true | residualDataFollowUp=true | wetLabAllowed=false | probabilityReported=false
Publication data fuel v3.0.6.9: route=APM_COMPOSITE_B2M_TAP_HLA | state=APM_TIER1_TIER2_DATA_FUEL_INJECTED_DATA_ONLY_FOLLOW_UP | syntheticData=false | fakeReplication=false | defense=APM is the only data-investable axis because of convergence, not because current MYCN-adjusted proof is publication-grade. | APM multi-cohort fuel v3.0.6.9.6: state=APM_MULTI_COHORT_DATA_FUEL_INJECTED_VALIDATION_TARGETS_ONLY | syntheticData=false | fakeReplication=false | targets=medulloblastoma, Wilms tumor/nephroblastoma, Ewing sarcoma | boundary=data-fuel only, not replication.

Pediatric cross-tumor scope v3.0.6.9.6: state=NEUROBLASTOMA_ONLY_SINGLE_COHORT_SCOPE | crossTumorPromotionAllowed=false | wetLabPromotionAllowed=false
Required external pediatric tumors: medulloblastoma,Wilms tumor / nephroblastoma,Ewing sarcoma

Investment triage v3.0.6.6: verdict=INVEST_IN_DATA_NOT_LAB | killedClaim=independent_MYCN_adjusted_APM_driver_claim | basis=invest because APM is a convergent downstream axis, not because the independent MYCN-adjusted claim is proven; kill=3, invest=2 | next=Invest only in data: full GPL5175 APM mapping, vector binding, GSE49710/TARGET-NBL direction replication, MYCN interaction/stability checks. No wet-lab yet. | probabilityReported=false
Question understood: true
Question: In pediatric neuroblastoma GSE85047, a composite antigen presentation score — defined by the concordant expression of B2M, TAP1, TAP2, and HLA-A — independently predicts progression-free survival after MYCN adjustment. Low composite antigen

presentation score identifies the immune-cold phenotype that AI-based pan-cancer models (Compass) identify as the primary determinant of immunotherapy non-response
Question understanding v3.0.3.97: APM_COMPOSITE_SCORE_AXIS | confidence=0.95
Mechanistic text understanding v3.0.4.3: COMPASS_APM_COMPOSITE_CONCEPT_TEXT | confidence=0.96
Text source context: Compass pan-cancer concept-model style text | prior=PAN_CANCER_MODEL_PRIOR_APM_CONCEPT_STRONG
Text genes understood: B2M,HLA-A,TAP1,TAP2,TAPBP,HLA-B,HLA-C
Mechanism claim parsed: A composite antigen-presentation machinery score may be a stronger immune-response concept than single markers.
Local proxy tests: APM_COMPOSITE_SCORE:DIRECT_LOCAL_EXPRESSION_SCORE_IF_PROBES_MAPPED
Not directly testable here: immunotherapy response endpoint,Compass 44-concept model inference,pan-cancer model transfer proof
Why this route: Compass,B2M,TAP1,TAP2,HLA
Cancer genes understood: B2M,TAP1,TAP2,HLA-A,MYCN,TAPBP,HLA-B,HLA-C
Dataset/outcome resolved: GSE85047 | PFS | covariates=MYCN
Literature axis basis: B2M/TAP/HLA/TAPBP composite is treated as a direct expression-survival concept score when probes are mapped.
Hypothesis tested: APM_COMPOSITE_PFS_MYCN_GSE85047
Primary axis: B2M_MHCI_NK_ESCAPE | route=PRIMARY_AXIS_APM_COMPOSITE_TEXT_SCORE
Genes understood: B2M,TAP1,TAP2,HLA-A,MYCN,TAPBP,HLA-B,HLA-C
Dataset/model: GSE85047 | outcome=PFS | model=APM composite low relaxed MYCN-aware screen (B2M/TAP1/HLA-A selected-matrix proxy; TAP2/TAPBP/HLA-B/HLA-C require full mapping)
Initial result: INITIAL_EXPLORATORY_ADVERSE_SIGNAL_REVIEW_ONLY | priority=SECONDARY_LEAD_REVIEW_ONLY
Triage signal: REVIEW_ONLY_DIRECTIONAL_SCREEN | level=REVIEW_ONLY_DIRECTIONAL_SCREEN_CANDIDATE | verdict=REVIEW_ONLY_COMPUTATIONAL_ROUND_OFFICIAL_CLAIM_BLOCKED
Signal promotion: rawAllowed=true | forKilledClaimAllowed=false | residualDataFollowUpAllowed=true | nextComputationalRoundCandidate=true | wetLabCandidate=false | officialClaimAllowed=false
Directional decision v3.0.4.1: STRONG_EXPLORATORY_DIRECTION | local=PRIORITIZE_AS_STRONG_LOCAL_DIRECTION | rank=1
Best direction scan: APM_COMPOSITE_PFS_MYCN_GSE85047 | STRONG_EXPLORATORY_DIRECTION | score=12.3647
Directional ranking table: #1 APM composite:STRONG_EXPLORATORY_DIRECTION | #2 B2M/TAP1/NK:STRONG_EXPLORATORY_DIRECTION | #3 CK2/APM fallback:APM_FALLBACK_SIGNAL_MECHANISM_NOT_DIRECTLY_TESTED | #4 HDAC/APM fallback:APM_FALLBACK_SIGNAL_MECHANISM_NOT_DIRECTLY_TESTED | #5 NK effector:DIRECTIONAL_LEAD_CONFOUNDER_SENSITIVE | #6 γδ/NKG2D:GAMMA_DELTA_SURROGATE_DIRECTION_TRGC_TRDV_FULL_MAPPING_REQUIRED | #7 CD276/B7-H3:SUBGROUP_ONLY_DIRECTION
Lead label: PRIMARY_DIRECTIONAL_LEAD
Directional why: Adverse local direction persists in MYCN-aware screen: delta=0.191176, rawHR=1.4159/p=0.0124, MYCN-adjusted HR=1.1971/p=0.246.
Directional next action: Run full GPL5175 mapping, source-certified tile build, MYCN interaction Cox, and independent replication.
Proxy attribution v3.0.4.4: DIRECT_AXIS_EXECUTED | signalSource=APM_COMPOSITE_DIRECT_PROXY | mechanismGeneTested=true | label=DIRECT_AXIS_EXECUTED
Probe lookup v3.0.5.0: resolvedVector=8 | resolvedButVectorMissing=20 | probeUnknown=0 |

absentOrReporterUnavailable=2 |
staticManifestLocalResolve=TARGETED_STATIC_MANIFEST_ATTACHED | runtimeNetwork=false |
legacyAutoFetchFlag=true
Probe vector binding v3.0.5.1: columnIndexPresent=true | resolvedProbeIds=28 |
boundVectors=8 | resolvedButColumnMissing=20 | aliasBoundVectors=0 |
nextBlocker=PARTIAL_BINDING_REMAINING_GENES_NEED_COLUMN_EXTRACTION
Evidence fuel v3.0.6.0: APM_COMPOSITE_B2M_TAP_HLA |
APM_COMPOSITE_B2M_TAP_HLA_FUEL_AVAILABLE_REVIEW_ONLY |
expected=STRONG_LOCAL_DIRECTION_BUT_TIER2_BLOCKED |
boundary=DATA_FUEL_IS_TEST_RECIPE_AND_SELECTED_RUNTIME_COVERAGE_NOT_TIER2_PROOF
Executed local directions v3.0.4.4: #1 APM
composite:STRONG_EXPLORATORY_DIRECTION[APM_COMPOSITE_DIRECT_PROXY] | #2
B2M/TAP1/NK:STRONG_EXPLORATORY_DIRECTION[B2M_TAP1_NK_DIRECT_PROXY] | #5 NK
effector:DIRECTIONAL_LEAD_CONFOUNDER_SENSITIVE[DIRECT_OR_ROUTE_SPECIFIC_SELECTED_MATRIX_PROX
| #6
γδ/NKG2D:GAMMA_DELTA_SURROGATE_DIRECTION_TRGC_TRDV_FULL_MAPPING_REQUIRED[GAMMA_DELTA_S
| #7 CD276/B7-H3:SUBGROUP_ONLY_DIRECTION[DIRECT_OR_ROUTE_SPECIFIC_SELECTED_MATRIX_PROXY] |
#8
ODC1/polyamine:WEAK_EXPLORATORY_DIRECTION[DIRECT_OR_ROUTE_SPECIFIC_SELECTED_MATRIX_PROXY]
Mechanistic hypotheses needing assets v3.0.4.4: CK2/APM
fallback:APM_FALLBACK_STRONG_SIGNAL_CK2_MECHANISM_NOT_DIRECTLY_TESTED | HDAC/APM
fallback:APM_FALLBACK_STRONG_SIGNAL_EPIGENETIC_MECHANISM_NOT_DIRECTLY_TESTED |
γδ/NKG2D:GAMMA_DELTA_SURROGATE_DIRECTION_TRGC_TRDV_FULL_MAPPING_REQUIRED |
NECTIN2/TIGIT:NECTIN2_TIGIT_NOT_TESTABLE_PROBE_AND_SINGLE_CELL_CONTEXT_REQUIRED
N=272 | events=96 | highEventRate=0.448529 | lowEventRate=0.257353 | delta=0.191176
Cox first-pass: state=RELAXED_COX_FIRST_PASS_EXECUTED_REVIEW_ONLY | HR=1.4159 | p=0.0124 |
rows=272 | events=96
MYCN-adjusted Cox first-pass: state=RELAXED_COX_FIRST_PASS_EXECUTED_REVIEW_ONLY |
HR=1.1971 | p=0.246 | rows=272 | events=96
MYCN-stratified event-rate: nonMYCN delta=0.077621 n=218 | MYCN-amp delta=0.242026 n=54
Component separation: antigenLoss delta=0.191176 | nkLow delta=0.352941 | combined
delta=0.191176
Tier-1 package v3.0.4.2: TIER1_FULL_GPL5175_LOCAL_COX_TARGET |
current=TIER0_DIRECTIONAL_SELECTED_MATRIX_ACTIVE | noStrictnessIncrease=true
GPL5175 annotation: FULL_GPL5175_ANNOTATION_REQUIRED_BEFORE_TIER1_PASS | fullPass=false
Verified-local Cox package: VERIFIED_LOCAL_COX_RERUN_CANDIDATE_RELAXED_MODE |
label=SECONDARY_AXIS_VERIFIED_LOCAL_COX_RERUN_TARGET_STRONG_EXPLORATORY_DIRECTION
Reproducibility receipt:
sha256-694025f7f14bf5da5ad2fbec5aaa7a283152de5b17fa696222d50e39826a5a2c |
replay=REPLAY_PACKAGE_SPEC_READY_CURRENT_RUN_EMBEDDED_JSON_ONLY
Method claim: The app parses cancer hypotheses, maps them to expression-survival axes, runs relaxed first-pass event-rate/Cox screens, ranks directional signals, and packages reproducibility receipts while blocking official proof/kill claims before Tier-2.
Biology claim: This axis is interpreted only as a local directional result in the current selected-matrix screen.
Evidence readiness gate: BLOCKED_BY_MISSING_EVIDENCE |
missing=official_or_declared_probe_platform_mapping,probe_to_gene_mapping_receipt,independent_replication_c
Evidence readiness receipts: platform/probe=PARTIAL_MAPPING_REVIEW_ONLY |
outcomeResolved=true | covariateResolved=true |
phenotypeComparator=SURVIVAL_OUTCOME_MODEL_NO_PHENOTYPE_COMPARATOR_REQUIRED_OR_NOT_DEC

| replicationComparator=REPLICATION_COMPARATOR_NOT_DECLARED |
confounder=CONFOUNDER_ADJUSTED_MODEL_EXECUTED_REVIEW_ONLY | mycnCoxExecuted=true |
fragility=RAW_SIGNAL_STRONG_BUT_WEAKENED_AFTER_MYCN_ADJUSTMENT_REVIEW_ONLY
Axis continuation: AXIS_FOLLOW_UP_REQUIRED_NOT_NEW_DISCOVERY | next=Do not stop at promising. Run axis transfer, direction-stability, known-driver and bulk-attribution checks using the supplied axis only.
Operational readiness: BLOCKED_FROM_PROMOTION_BY_READINESS_GAPS — Resolve first: official_or_declared_probe_platform_mapping, probe_to_gene_mapping_receipt, independent_replication_or_cohort_transfer_check
Final hypothesis state v3.0.6.0: STRONG_LOCAL_DIRECTION_BUT_TIER2_BLOCKED | reason=APM composite shows a strong local selected-matrix direction, but Tier-2 computational evidence is blocked until full APM mapping, executable statistics receipt, and independent replication.
Decisive adjudication v3.0.6.5:
verdict=KILLED_INDEPENDENT_MYCN_ADJUSTED_CLAIM_LOCAL_DIRECTION_SURVIVES | claimKilled=true | localSignalSurvives=true | notTestable=false | action=Stop mechanism claims; keep only the local APM direction for replication or full mapping.
Investment verdict v3.0.6.6: verdict=INVEST_IN_DATA_NOT_LAB |
killedClaim=independent_MYCN_adjusted_APM_driver_claim | killCriteria=3 | investCriteria=2 | next=Invest only in data: full GPL5175 APM mapping, vector binding, GSE49710/TARGET-NBL direction replication, MYCN interaction/stability checks. No wet-lab yet.
FINAL INVESTMENT DECISION v3.0.6.7: INVEST_IN_DATA_ONLY | claim=current independent/proof claim killed; downstream APM data question remains investable |
killedClaim=independent_MYCN_adjusted_APM_driver_claim | criteria=kill:3/invest:2 |
next=Invest only in data: full GPL5175 APM mapping, vector binding, GSE49710/TARGET-NBL direction replication, MYCN interaction/stability checks. No wet-lab yet. |
probabilityReported=false
PREPRINT READINESS v3.0.6.8: PREPRINT_READY_WITH_DATA_ONLY_FOLLOW_UP | title=Hypothesis Stress-Test & Investment Decision Report v3.0.6.8 | killedAsWritten=true |
residualDataFollowUp=true | wetLabAllowed=false | probabilityReported=false
Publication data fuel v3.0.6.9: route=APM_COMPOSITE_B2M_TAP_HLA |
state=APM_TIER1_TIER2_DATA_FUEL_INJECTED_DATA_ONLY_FOLLOW_UP | syntheticData=false |
fakeReplication=false | defense=APM is the only data-investable axis because of convergence, not because current MYCN-adjusted proof is publication-grade. | APM multi-cohort fuel v3.0.6.9.6:
state=APM_MULTI_COHORT_DATA_FUEL_INJECTED_VALIDATION_TARGETS_ONLY | syntheticData=false |
fakeReplication=false | targets=medulloblastoma, Wilms tumor/nephroblastoma, Ewing sarcoma | boundary=data-fuel only, not replication.
Pediatric cross-tumor scope v3.0.6.9.6: current=NEUROBLASTOMA_ONLY_SINGLE_COHORT_SCOPE |
observed=neuroblastoma | crossTumorPromotionAllowed=false | wetLabPromotionAllowed=false
Pediatric expansion targets: medulloblastoma,Wilms tumor / nephroblastoma,Ewing sarcoma
Pediatric scope next action: After neuroblastoma, run the same stress-test template on medulloblastoma, Wilms tumor/nephroblastoma, and Ewing sarcoma; keep each as KILLED / NULL-LIKE / SURVIVING / NOT_MEASURABLE with separate receipts before any cross-tumor claim.
Pediatric scope boundary: Neuroblastoma-only evidence remains single-cohort and disease-scoped. Cross-pediatric-tumor credibility requires independent human datasets from at least three additional pediatric tumor families before any generalization, Tier-2 promotion, or wet-lab language.
Preprint boundary: This report supports a methods/resource-allocation claim only: supplied

computational hypotheses are triaged as killed-as-written, watchlist-only, or data-only follow-up. It does not prove or disprove biology, does not recommend treatment, and does not establish clinical actionability.
Allowed manuscript sentence: APM/B2M-TAP-HLA remains a data-only follow-up axis because independent upstream hypotheses converge on antigen-presentation loss, not because the current MYCN-adjusted model proves an independent biology claim.
Direction: HIGH_SCORE_WORSE_PFS_EVENT_RATE | caveat=Relaxed first-pass screen only. Shows trial event-rate, MYCN-stratified and in-app Cox-review numbers for debugging. Not proof, not official support/kill, not clinical, not publication-grade, and not Tier-2 computational evidence.
Missing gates: official TAP2/TAPBP/HLA-B/HLA-C full GPL5175 mapping,Compass/immunotherapy response endpoint not locally available,full GPL5175 audit,source certification,independent Cox-capable replication,Tier-2 not reached,full GPL5175 mapping required before official support/kill,independent replication required before computational evidence language
Claim guard: BLOCKED_COMPUTATIONAL_CLAIM_REQUIRES_TIER2 | tier=TIER0_SLICE_REPRODUCIBLE_REVIEW_ONLY | officialPromotionAllowed=false | experimentalPromotionAllowed=false | experimentalPromotionForKilledClaimAllowed=false | dataFollowUpAllowed=true | computationalReplicationAllowed=true | wetLabAllowed=false | officialClaimAllowed=false
Tier-0 proves replayability, not source truth.
Tier-1 proves source extraction, not biological truth.
Tier-2 is required before any computational evidence claim.